

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Concept Mapping Questions

Course Code:	CSA0315	Course Title:	Data Structures
CO Assessed:	C01	Assessment:	Assessment Tool 1 – Concept Mapping Questions
Weightage:	10% of CIA	Total Marks:	100
C01 Statement:	Basics, Data, Data object, Data structure, Abstract Data Types (ADT), realization of ADT in 'C', Primitive and non-primitive data structure, Linear and non-linear data structure Arrays & Recursion, Performance analysis and Measurement: Time and space analysis of algorithms, Average, best and worst-case analysis.		
Student Name:	R.Usha	Reg. No.:	192525100
Section / Batch:	2025	Date:	14-07-26

Code of Conduct

I R.Usha (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: R.Usha

Instructions

- This test contains 3 Concept mapping questions. Total: 100 Marks.
- When a student answer using a concept map, in evaluation the answer should contain following four requirements: Understanding of concepts, Connections between ideas, Logical structure, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

Section / Topic	Focus	Q Nos	Marks
Basics, Data, Data object, Data structure, Abstract Data Types (ADT), realization of ADT in 'C'	Concept mapping of Basics in Data Structure	1	30
Performance analysis and Measurement: Time and space analysis of algorithms, Average, best and worst-case analysis.	Fibonacci series	2	40
Primitive and non-primitive data structure, Linear and non-linear data structure Arrays	Concept mapping on classifications of data structures.	3	40

Annexure A – Application Based Assessment

- 1) Write a c program and draw the flow chart as concept map for finding the count of odd and even numbers in an array of n numbers.
- 2) Write a concept map based flow chart and also write the program to generate the Fibonacci series and sum of the series up to n numbers.
- 3) A smart city traffic management system handles data related to vehicle movement, road networks, and signal control. Construct a concept map connecting Primitive Data Structures → Non-Primitive Data Structures → Linear → Non-Linear Structures. Include examples such as arrays, linked lists, trees, and graphs, and illustrate their applications in traffic management. Explain the relationships between these structures and justify their classification based on functionality and efficiency.

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution

Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand
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Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Date
14/12/20
Tuesday

CSA0315 - Data Structures

Assignment - CO1-ATI

Name: R. Usha
Reg no: 192525100

Q) C program to Count odd and even elements in an array

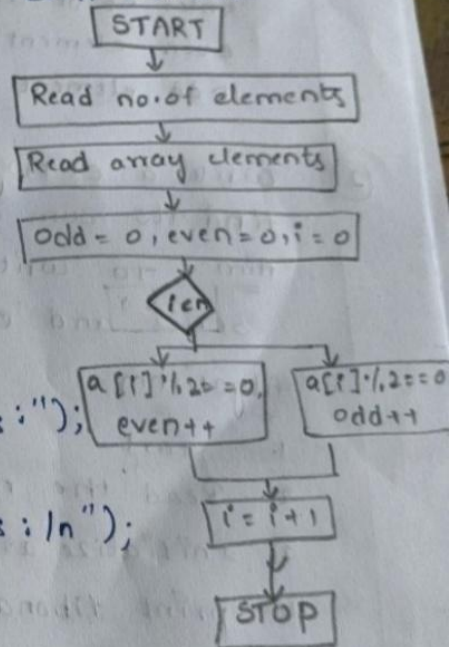
Aim: TO write a C program to count the number of odd and even elements (n)

Algorithm

1. start
2. Read the number of elements (n)
3. Read the array element
4. initialize odd=0 and even=0
5. check each element
6. Display the no. of odd and even element.
7. stop.

Program

```
#include <stdio.h>
int main ()
{
    int n, i, a[100];
    int odd = 0, even = 0;
    printf ("enter no. of elements:");
    scanf ("%d", &n);
    printf ("enter array elements: \n");
    for (i=0; i<n; i++)
    {
        scanf ("%d", &a[i]);
    }
```



```

for (i=0; i<n; i++)
{
    if (a[i] % 2 == 0)
        even++;
    else
        odd++;
}
printf ("even elements = %d\n", even);
printf ("odd elements = %d\n", odd);
return 0;
}

```

output:

enter no. of elements = 5

enter no. of array element = 10 15 8 7 20

even element = 3

odd element = 2

② Write a c-program to print fibonacci series and find its sum of series.

Aim:- To write a c program to print the fibonacci series and calculate its sum.

Algorithm:-

1. start
2. Read the no. of term
3. Initialize $a=0$, $b=1$, $sum=0$
4. print fibonacci term
5. Add each term to sum
6. Display the total sum
7. stop.

Program:-

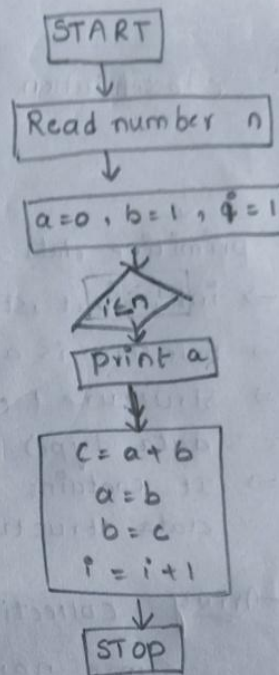
```
#include <stdio.h>
int main()
{
    int n, i;
    int a = 0, b = 1, c;
    int sum = 0;
    printf("enter no. of terms:");
    scanf("%d", &n);
    printf("fibonacci series:\n");
    for (i = 1; i <= n; i++)
    {
        printf("%d", a);
        sum = sum + a;
        c = a + b;
        a = b;
        b = c;
    }
    printf("sum = %d", sum);
    return 0;
}
```

output

enter no. of elements = 7

fibonacci series : 0 1 1 2 3 5 8

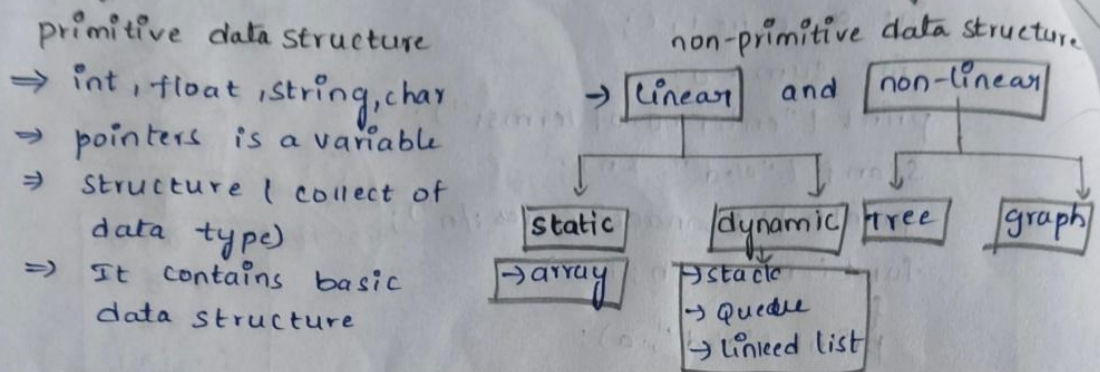
Sum = 20



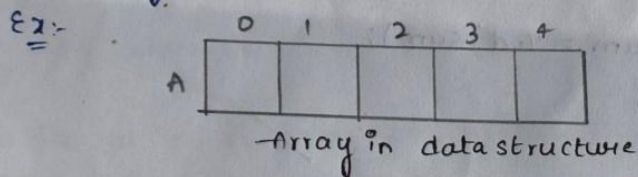
③ concept map for data structure and classification

Defination:- A data structure is a method of organizing the data from the memory so that it can be accessed and modified efficiently.

Classification of data structure

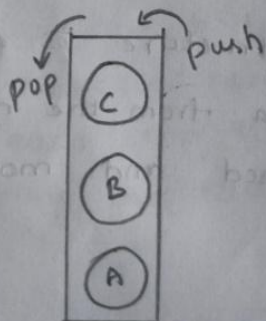


Array:- Collection of location in memory under the same name and same data type is called array.



Stack:- last in first out (The boundary of data structure is fixed)

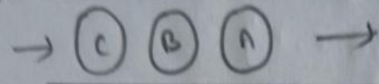
non-stack:- no boundary & and memory locations are not fixed.



stack of data structure (LIFO)

Queue:-

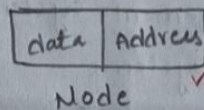
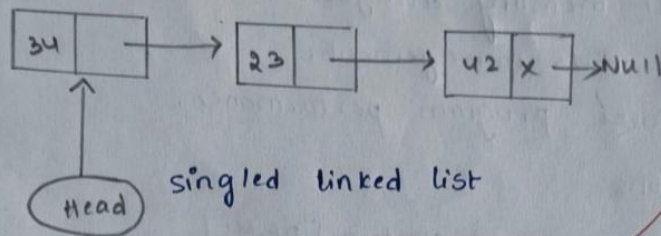
First in first order or mannaylo.
enqueue dequeue



Queue data Structure (FIFO)

Linked list:-

Connected to one data to another data
with the help of address.



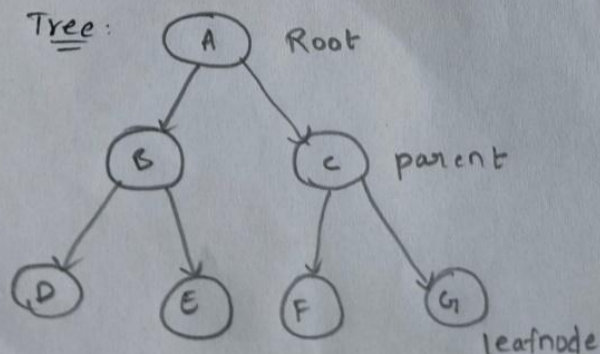
Non-linear data structure:-

⇒ It is a non-sequential order.

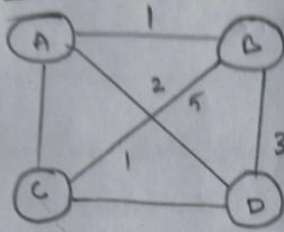
⇒ These are classified into 2 types.

1. Tree data structure
2. Graph data structure.

Tree:-



Graph:-



$$G = \{V, E\}$$

$$V = \{A, B, C, D\}$$

$$E = \{AB, AC, BC, CD, AD\}$$

Advantages:-

- efficient storage of data
- Faster Searching and sorting.
- Better memory utilization.
- Easy data management.
- improves program performance.

Applications:-

- ⇒ Data management system [DBMS]
- ⇒ Operating systems
- ⇒ computer network
- ⇒ Artificial intelligence
- ⇒ software development.